Response to the Reviewers

Bayesian Adaptive Portfolio Optimisation Exploiting Fully Probabilistic Design

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Marking. In the revised manuscript, blue marks what changed with respect to the version the reviewers saw. Subsections that are entirely new carry a blue title and black body, so that the new experimental section stays readable; rewritten passages of existing sections are blue in full. The few blue marks of the previous round have been cleared. The Complementary Material is new in its entirety and is therefore not marked.

We thank both reviewers for their careful reading. The experimental section has been rebuilt from scratch around the reviewers' requests and, to keep the paper within twelve two-column pages, its details are reported in an accompanying Complementary Material document, whose sections, figures and tables carry the prefix S, and in doing so we found and corrected an implementation error that affected all previously reported numbers. We describe the correction first because it changes the conclusions, then answer point by point. All theory-related points (covariance scaling, the ω optimisation, Eq. (17), the multiplier update, the contribution statement) are addressed separately by the second author [MK: fill].

What changed and why

1. **The constrained optimiser did not optimise.** The inner projected-gradient loop of the SMALBE implementation reset its iterate to the outer iterate on every iteration, so the routine returned its starting point, the softmax of the unconstrained LQR action. The tables of the submitted manuscript therefore described this heuristic, whose weights stay within a few percent of uniform. The routine is fixed (it agrees with an independent SLSQP solver to 10^{-11} on random instances; test included in the released code) and the revised paper reports both the corrected policy and, clearly labelled, the legacy heuristic.

2. **The risk weight ω^* is always the boundary of its search interval.** Reviewer 2 anticipated this from the first-order condition, and it is correct on three counts. (i) The inner minimum in Eq. (20) is minus the logarithm of a Gaussian integral that decreases monotonically in ω, so the outer maximisation has no finite solution; the printed second line of the derivation also has the opposite sign of the first. (ii) The necessary condition is a sum of two positive terms and has no root; the implementation minimised its left-hand side on $[10^{-8}, 10^{3}]$ and returned the upper bound in 100% of the 10^5 decision steps of the study. (iii) With $\omega = 10^3$ the variance term dominates the loss by four orders of magnitude and the cost term is absent, which is why the corrected policy trades 15% of the portfolio per day. Sec. S7 of the complementary material caps ω and tries three principled elicitations: any ω at which the return forecasts influence the allocation loses most of the capital, because the forecasts are as large as the returns (standard deviation 0.027 vs. 0.027) and carry no information (correlation with the realised returns -0.01, sign accuracy 49.7%, squared error twice that of predicting zero). The derivation of ω^* is withdrawn from the theoretical claims and the quantity is treated as a user-set bound; the theory section is revised accordingly [MK].

3. **Numerical fixes.** Determinants of the 130-plus-dimensional statistics under/overflow in double precision inside the forgetting and structure likelihoods; they are replaced by log-determinants. The prior term of the structure likelihood used the posterior block $\mathbf{V}_y$ instead

of $\mathbf{V}_{y,0}$. After structure estimation the prior square-root factor is now recomputed from the reduced $\mathbf{V}_0$.

4. **The empirical conclusions are different.** Under a common cost model the corrected policy has a smaller maximum drawdown than every non-min-variance benchmark but a lower Sharpe ratio than the equal-weight portfolio, because of turnover. The legacy heuristic is statistically indistinguishable from the rebalanced equal-weight portfolio. The paper no longer claims outperformance of the uniform portfolio or of the index; it presents the framework, the corrected empirical picture, the ablation that shows which components matter, and the two design elements (ω elicitation and forgetting prior) that the experiments identify as the ones to change.

Reviewer 9

First, the evidence of practical outperformance is not fully convincing . . . Why not comparing the SP500 also in the 500 days period?

Response. The S&P 500 (SPY, total return) is now reported on the full 500-day test period, on the three additional windows of Sec. S9, and in every paired table. It beats every strategy on the 20-stock universe, including equal-weight buy and hold, on this period; we say so and no longer claim to beat the index. The live test is kept as an anecdote, with the caveat that 32 trading days cannot separate strategies; the same period is covered with 100 portfolios by the 2024–2026 out-of-sample window.

Second, I recommend adding stronger and more standard benchmarks . . .

Response. Sec. 4.1 of the paper and Sec. S3 of the complementary material now compare, on identical portfolios, days and cost model: equal-weight rebalanced, equal-weight buy and hold, inverse volatility, minimum variance (120 and 250 days, Ledoit–Wolf), mean-variance (120, 250 days, and a cost-aware version), an ARX-conditional mean-variance that uses our regressors with a classical estimator and optimiser, and the S&P 500. All differences are tested with paired Wilcoxon tests, bootstrap confidence intervals and win rates over 200 portfolios.

Third, I recommend publish a open-source version of the code.

Response. The complete code, data-preparation scripts, the experiment harness that regenerates every table and figure of Sec. 4 from one command, and the solver test are released at `https://github.com/ProchTomas/ResearchProject` [Zenodo DOI].

I think that the results obtained do not justify the use of such a complex method since it does not have a clear outperformance compared to simple portfolios . . . why do not try it in another domain?

Response. We agree that the previous evidence did not justify the claims, and the corrected evidence justifies them even less; the revised paper says this explicitly. What the new experiments do establish is (i) which components of the pipeline are necessary for the daily forecasts to be usable at all (the FPD risk term, the structure estimation, the point estimate: removing any of them loses half of the capital), (ii) that the data-based forgetting rejects outliers as designed, verified on synthetic data with known parameters, and (iii) that the two remaining weaknesses are localised: the ω elicitation, which saturates, and the strength of the optimal prior, which together freeze the model at its initial estimate. We consider a controlled comparison in which a complex method is shown not to beat simple benchmarks, with the reasons identified, to be a useful result for the community, and we have rewritten the abstract, highlights and conclusions accordingly. The synthetic experiment of Sec. S8 is a first step towards the less noisy domain the reviewer suggests.

Reviewer (attachment 416624)

Major 3: The proposed strategy is mainly compared against two equal-weight portfolios . . . More competitive baselines and/or ablation studies are needed . . .

Response. See the benchmark list above. Sec. 4.3, with its tables in Sec. S5, adds an ablation on 100 portfolios in which one component at a time is removed or replaced: forgetting (none / exponential), the FPD risk term, structure estimation, the optimal prior, the receding horizon and loss-to-go propagation, Thompson sampling, and the optimiser (constrained QP vs. the legacy heuristic). Each is compared to the full policy with paired tests.

In Table 2, the Sharpe ratios appear inconsistent with the definition . . .

Response. The reviewer is right; the entry was a transcription error. All tables are now generated directly from the result files by the released script, and the Sharpe ratio is reported annualised.

Table 3 shows that the proposed method underperforms the transaction-cost equal-weight portfolio across all reported metrics in the live test, and does not improve drawdown relative to the S&P 500. These results should be discussed more explicitly.

Response. They are, in Sec. S10, together with the 2024–2026 window that covers the live-test period with 100 portfolios instead of one.

Finally, the claimed robustness to non-Gaussian behaviour and "black swan" events is not sufficiently supported, as no dedicated theoretical analysis or stress-test experiments are provided.

Response. Sec. S8 adds a synthetic stress test (ARX data with known parameters, one-day crashes on all assets, an abrupt change of all coefficients) comparing the data-based forgetting with no forgetting and exponential forgetting, and a real-data view of the forgetting weights and drawdowns around the April 2025 tariff shock, plus the COVID-2020 and 2022 bear-market windows. The forgetting rejects the outlying observations exactly as designed. The same experiments show that it does not track parameter drift and that it discards most ordinary observations too; we report both. Robustness claims in the abstract and Sec. 2.3 are restricted to what is shown.

Major 4: . . . the complete framework still involves a number of important user-specified choices . . . an end-to-end pseudocode or algorithm box would substantially improve clarity and reproducibility.

Response. An end-to-end algorithm box and a table of all user-set quantities, with their values and how they were chosen, are added as Sec. S4 of the complementary material. Sec. S6 reports the sensitivity to μ, (α_0, β_0) and $\omega_{\max}$ on 100 portfolios. The claim of "minimal tuning" is reworded: the number of knobs is small, but two of them (μ and the forgetting prior) matter a great deal.

Minor 3: The experimental setup should be documented more clearly . . . potential selection bias associated with screening assets based partly on historical performance.

Response. Sec. S1 now gives the data source, the ticker list, the exact dates of every phase, the cost model with its formula, the portfolio sampling and the seeds. The screen on 2021 returns precedes the test period but is a screen on past performance; the additional windows use the same universe on periods before the screen and after the paper's data, and the comparison with the S&P 500 shows that the screened universe underperformed the index in 2023–2025.

Minor 1, 2: typographical issues; statements stronger than warranted ("silver bullet" . . .).

Response. Corrected (Throughout, joint, Sharpe, Riccati; equation numbering); "silver bullet" and the black-swan wording are removed.